Rahul Rawat

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Siemens Healthineers
Hiring Team

Application: Working Student Data Science and AI for X-Ray Technology in Forchheim

Dear Hiring Team,

I am applying for the Working Student position on Data Science and AI for X-Ray Technology at Siemens Healthineers in Forchheim. The brief on supporting the development and evaluation of AI based methods for analysing unstructured data such as images of technical components, working hands on in Python, and exploring and applying deep learning models, aligns directly with the Python ML delivery and honest model evaluation work I have been shipping over the past year.

At eRay GmbH I built an end to end recursive time series pipeline forecasting four water quality indicators on real sensor data, benchmarking six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. I enforced strict anti leakage rules and surfaced the honest finding that some indicators are physically predictable while others are not without live optical sensors. That leakage aware, evaluation first mindset carries directly into X-Ray image and technical component analysis, where honest evaluation is the difference between a model that ships and one that fails silently.

In CreditIQ I lifted the model's Disparate Impact ratio from 0.79 to 0.88, cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, and backed the pipeline with unit tests at 100 percent branch coverage. My bachelor thesis compares six classifiers on a 768 patient clinical dataset with 10 fold cross validation, catches biologically impossible zero values that the original authors had overlooked, and chooses ROC AUC over accuracy on a 65 to 35 imbalanced target to avoid a misleadingly rosy accuracy figure. That is exactly the disciplined, healthcare adjacent evaluation Siemens Healthineers needs.

I am proficient in Python and scikit learn, comfortable with deep learning frameworks, and I speak English fluently with German at B1 in progress. I hold the NVIDIA Building LLM Applications With Prompt Engineering, AWS Academy Cloud Foundations, and Google Data Analytics Foundations certificates. I am available for the fixed term 15 hours per week Working Student position and would be glad to align the exact scope with the X-Ray Technology team in Forchheim.

Kind regards,

Rahul Rawat